Airlines Pricing

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The purpose of this project is to analyze and predict airline tickets prices using data mining learning techniques. Airline ticket pricing is an important issue for both businesses and consumers. Prices can vary depending on factors such as travel class, departure, and how early a ticket is purchased. As someone who likes to travel and is self-cautious on financial decisions, this topic is relevant because it shows how dynamic pricing affects everyday consumers. No matter if its commercial or private, dynamic pricing has affected customers for the benefit of the company or shareholders.

From a business perspective, airlines aim to maximize revenue by adjusting prices based on demand, timing, and customer behavior. At the same time, consumers want to find the most affordable tickets and understand when and how prices change. This creates a strong need for predictive models that can estimate ticket prices based on key variables. Stakeholders in the problem in this problem include airline companies, travel agencies, and consumers from a better understanding of pricing patterns.

This project focuses on predicting airlines ticket prices using a regression-based approach. By analyzing features such as airlines, source and destination cities, number of stops, travel class, duration, and days left until departure, the goal is to identify which factors most influence pricing and build models that can accurately estimate ticket prices. The results of this analysis can provide insights into pricing strategies and help improve decision-making for both businesses and travelers.

The problem addressed in this project is the difficulty of predicting airline ticket prices due to the number of factors influencing pricing. Airline tickets prices are not fixed and can vary based on several variables such as the airline, departure and arrival locations, travel class, duration of the flight, number of stops, and how far in advance the ticket is purchased. It is challenging for both businesses and consumers to understand how prices are determined and decided.

The primary analytical question explored in this project is: Can airlines ticket prices be accurately predicted using available flight-related features? Some questions can be: Which variables have the greatest impact on ticket pricing? And Which machine learning provide the most accurate prediction? These questions are important because they help identify patterns in pricing and evaluate the effectiveness of predictive modeling techniques.

To address these questions, a regression-based approach is the best way to answer these questions. Regression is the correct choice because the target variable, ticket price, is continuous. Multiple models are implemented, including Linear Regression, Decision Tree Regression, and K-Nearest Neighbors (KNN), to compare their performance. By evaluating these models using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2), the project determines which method best captures the relationship between the input features and airline ticket prices.

The dataset used in this project was the Airlines Flights Data dataset obtained from Kaggle. The original dataset contained approximately 300,153 instances and 12 attributes related to airlines ticket pricing and information. For this project, a random sample of 1,200 observations was selected to improve computational efficiency while still maintaining enough data for predictive modeling.

The target variable of interest in this project in this project was price, which represents the cost of an airline ticket. Since ticket pricing is a continuous numerical variable, a regression-based machine learning approach was used. The remaining attributes were used as predictor variables to estimate tickets prices.

The dataset included both categorical and numerical attributes. The categorical variables included airlines, source city, departure time, stops, arrival time, destination city, and travel class. The numerical variables included flight duration and days left until departure. The index and flight columns

were removed during preprocessing because they did not contribute meaningful predictive information to the models.

Before model development, the categorical variables were transformed into numerical format using one-hot encoding with the get_dummies() function in pandas. This process allowed machine learning algorithms to properly intercept categorical data. The dataset was then divided into training and testing sets using an 80/20 split. Additionally, feature scaling using StandardScaler was applied before implementing model such as Linear Regression and K-Nearest Neighbors (KNN), since these algorithms are sensitive to differences in feature magnitudes.

One challenge with the dataset was handling multiple categorical variables and converting them into a usable numerical format for machine learning models. Another challenge involved selecting appropriate machine learning models and turning parameters to improve predictive performance. Despite these challenges, the dataset provided sufficient information and variability to support regression analysis and model comparison.

To begin the analysis, exploratory data analysis (EDA) techniques were used to better understand the dataset and identify important patterns related to related to airline ticket pricing. The dataset included both categorical and numerical variables, requiring preprocessing before machine learning models could be implemented. Summary statistics and features inspection were performed to better understand the distribution of ticket prices and the characteristics of the flight data.

Several preprocessing techniques were applied before building the predictive models. First, unnecessary attributes such as index and flight were removed because they did not provide meaningful predictive value. The categorical values were then converted into numerical variables using one-hot encoding through get_dummies() function in pandas. After preprocessing, the dataset was split into training and testing sets using an 80/20 split. Feature scaling with StandardScaler was also applied

because some machine learning algorithms, particularity K-Nearest Neighbors (KNN), are sensitive to differences in feature magnitudes.

Three regression models were implemented and compared in this project: Linear Regression, Decision Tree Regressor, and K-Nearest Neighbors (KNN) Regressor. These models were selected because they represent different machine learning approaches and allow for comparison between linear, tree-based, and distance-based predictive methods.

Linear Regression was used as a baseline model because it is one of the most common regression algorithms and works well for identifying linear relationships between variables. The model attempts to minimize prediction error by fitting a linear equation to the dataset. The Linear Regression model achieved an R-squared value of approximately 0.50, indicating that the model explained around 50% of the variation in airline ticket prices.

The Decision Tree Regressor model was implemented to capture more complex and non-linear relationships within the dataset. GridSearchCV was used to tune the model's hyperparameters, including criterion, maximum depth, minimum samples splits=, and minimum samples leaf. The best-performing configuration used the absolute_error criterion with a maximum depth of 10. The Decision Tree Model achieved the strongest overall performance with an R-squared value of approximately 0.56 and the lowest prediction error among the tested models. This suggests that the airline ticket pricing contains non-linear patterns that are better captured by tree-based methods.

The K-Nearest Neighbors (KNN) Regressor was also implemented and tested using multiple values of k to identify the best-performing neighbor size. The highest performance was achieved with k = 15. However, the KNN model produced the weakest results overall, with an R-squared value of approximately 0.33. This lowers performance may be due to the high dimension of the dataset and the limitations of distance-based methods when working with many encoded categories features.

Overall, the results showed that the Decision Tree Regressor performed best among the tested models. The model was more effective at capturing the complexity of airline pricing patterns compared to Linear Regression and KNN. The findings suggest that factors such as travel class, duration, departure timing, and days left until departure all contribute significantly to airline ticket pricing behavior.

By analyzing variable such as airline, source and destination cities, travel class, duration, number of stops, and days left until departure, the project successfully identified patterns that influence airline ticket pricing. Three different regression models were implemented and evaluated: Linear Regression, Decision Tree Regressor, and K-Nearest Neighbors (KNN) Regressor. The models were compared using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). Among the model tested, the Decision Tree Regressor achieved the best overall performance with the highest (R^2) value and lowest prediction error. This suggests that airline ticket pricing contains complex and non-linear relationships that are better capture by tree-based machine models.

The findings of this project demonstrate that machine learning can be effectively used to analyze and predict airlines ticket pricing. These results may help airlines improve pricing strategies while also helping consumers better understand how ticket pricing change based on different factors. Additionally, the project highlights the importance of preprocessing data and selecting appropriate machine learning models when performing predictive analytics.

Although the models produced meaningful results, there are still opportunities for future improvement. Additional research could also explore how external factors such as holidays, weather, for fuel prices influence airline ticket pricing behavior.

References

Grewal, R. (2025). Airlines flights data [Data set]. Kaggle.

https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data